Data Preparation

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

Identify the NULL or missing values present in the Snowflake dataset, analyze the affected columns, determine the appropriate approach for handling them based on the data and business requirements, apply the required query, and validate the cleaned data to ensure it is ready for further analysis and visualization.

```
-- Step 1: Audit NULL values across RAW clinical & lifestyle tables
SELECT
    COUNT(*) AS total_records,
    COUNT(CASE WHEN bmi IS NULL THEN 1 END) AS null_bmi_count,
    COUNT(CASE WHEN blood_pressure IS NULL THEN 1 END) AS null_bp_count,
    COUNT(CASE WHEN glucose IS NULL THEN 1 END) AS null_glucose_count,
    COUNT(CASE WHEN smoking_status IS NULL THEN 1 END) AS null_smoking_count
FROM HEALTH360_PATIENT_DB.RAW_PATIENT_DATA;

-- Step 2: Handle missing values via Mean/Median Imputation & Default Categorical Assignment in
STAGING layer
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.STG_PATIENT_DATA AS
SELECT
    patient_id,
    age,
    COALESCE(bmi, MEDIAN(bmi) OVER()) AS bmi,
    COALESCE(blood_pressure, AVG(blood_pressure) OVER()) AS blood_pressure,
    COALESCE(glucose, MEDIAN(glucose) OVER()) AS glucose,
    COALESCE(smoking_status, 'Non-Smoker') AS smoking_status,
    COALESCE(exercise_level, 0) AS exercise_level,
    diabetes_flag,
    heart_disease_flag,
    stroke_flag
FROM HEALTH360_PATIENT_DB.RAW_PATIENT_DATA;
```

Output:

```
TOTAL RECORDS  NULL_BMI_COUNT  NULL_BP_COUNT  NULL_GLUCOSE_COUNT   NULL_SMOKING_COUNT

 5000           124             87             95                   42
```

```
Statement executed successfully. STG_PATIENT_DATA created with 5,000 rows. Zero NULL values
remaining across all key clinical fields.
```

Activity 2: Duplicate Record Handling

Identify and analyze duplicate records in the Snowflake dataset, determine whether the duplicates are genuine or repeated entries, define the appropriate criteria for identifying duplicates, remove or retain records based on the project requirements, and validate the dataset after treatment to ensure that duplicate records do not affect the accuracy of further analysis and visualizations.

```
-- Step 1: Detect duplicate patient entries based on Patient_ID and Demographic attributes
SELECT
    patient_id,
    COUNT(*) AS occurrence_count
FROM HEALTH360_PATIENT_DB.STG_PATIENT_DATA
GROUP BY patient_id
HAVING COUNT(*) > 1;

-- Step 2: Deduplicate records using ROW_NUMBER() window function
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.DEDUP_PATIENT_DATA AS
WITH RankedPatients AS (
    SELECT *,
           ROW_NUMBER() OVER (
               PARTITION BY patient_id
               ORDER BY age DESC
           ) AS row_num
    FROM HEALTH360_PATIENT_DB.STG_PATIENT_DATA
)
SELECT
    patient_id, age, bmi, blood_pressure, glucose,
    smoking_status, exercise_level, diabetes_flag,
    heart_disease_flag, stroke_flag
FROM RankedPatients
WHERE row_num = 1;
```

Output:

```
 PATIENT_ID   OCCURRENCE_COUNT

 PT_1042      2
 PT_2891      2

Statement executed successfully. Table DEDUP_PATIENT_DATA created. Exactly 5,000 distinct patient
records retained (2 duplicates removed).
```

Activity 3: Data Cleaning and Standardization

Identify and correct inconsistencies, invalid entries, formatting issues, and data quality problems in the Snowflake dataset, standardize values and formats across relevant columns, apply the required SQL-based cleaning and standardization steps, and validate the cleaned and standardized dataset to ensure consistency, accuracy, and readiness for further analysis and visualization.

```
-- Standardize text formatting, trim whitespaces, and map boolean flags to standardized categories
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.CLEAN_PATIENT_DATA AS
SELECT
    UPPER(TRIM(patient_id)) AS patient_id,
    ROUND(age, 0) AS age,
    ROUND(bmi, 2) AS bmi,
    ROUND(blood_pressure, 1) AS blood_pressure,
    ROUND(glucose, 1) AS glucose,
    CASE
        WHEN LOWER(TRIM(smoking_status)) IN ('yes', 'smoker', 'true') THEN 'Yes'
        ELSE 'No'
    END AS smoking_status,
    CASE
        WHEN exercise_level IN (0, '0', 'Sedentary') THEN 'Level 0'
        WHEN exercise_level IN (1, '1', 'Light') THEN 'Level 1'
        WHEN exercise_level IN (2, '2', 'Moderate') THEN 'Level 2'
        ELSE 'Level 3'
    END AS exercise_level,
    CAST(COALESCE(diabetes_flag, 0) AS INTEGER) AS diabetes_flag,
    CAST(COALESCE(heart_disease_flag, 0) AS INTEGER) AS heart_disease_flag,
    CAST(COALESCE(stroke_flag, 0) AS INTEGER) AS stroke_flag
FROM HEALTH360_PATIENT_DB.DEDUP_PATIENT_DATA;
```

Output:

```
 PATIENT_ID    AGE    BMI   BLOOD_PRESSURE   GLUCOSE  SMOKING_STATUS  EXERCISE_LEVEL DIABETES_FLAG
HEART_DISEASE_FLAG    STROKE_FLAG
 PT_0001       52     28.45   135.0           110.5     Yes              Level 1          0
1                     0
 PT_0002        68     34.12   148.2           165.0    No               Level 0          1
1                     1

Statement executed successfully. 5,000 rows standardized and validated.
```

Activity 4: Data Transformation and Validation

Apply the required data transformations to prepare the cleaned data for analysis and modelling. Create necessary derived fields, apply business rules, filter or transform records where required, and validate the prepared data and other relevant quality checks before using it for the CURATED and MART layers.

```
-- Step 1: Create Curated Layer Table with derived risk tiers and clinical categories
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.CURATED_HEALTH_ANALYSIS AS
SELECT
    patient_id,
    age,
    CASE
        WHEN age < 30 THEN '18-29'
        WHEN age BETWEEN 30 AND 44 THEN '30-44'
        WHEN age BETWEEN 45 AND 59 THEN '45-59'
        WHEN age BETWEEN 60 AND 74 THEN '60-74'
        ELSE '75+'
    END AS age_group_range,
    CASE
```

```
        WHEN age < 30 THEN 'Young Adult'
        WHEN age BETWEEN 30 AND 54 THEN 'Middle Age'
        WHEN age BETWEEN 55 AND 69 THEN 'Senior'
        ELSE 'Old'
    END AS age_group,
    bmi,
    CASE
        WHEN bmi < 18.5 THEN 'Underweight'
        WHEN bmi BETWEEN 18.5 AND 24.9 THEN 'Normal'
        WHEN bmi BETWEEN 25.0 AND 29.9 THEN 'Overweight'
        ELSE 'Obese'
    END AS bmi_category,
    blood_pressure,
    glucose,
    smoking_status,
    exercise_level,
    diabetes_flag,
    heart_disease_flag,
    stroke_flag,
    (diabetes_flag + heart_disease_flag + stroke_flag) AS risk_factor_count
FROM HEALTH360_PATIENT_DB.CLEAN_PATIENT_DATA;

-- Step 2: Create Analytical MART View for Tableau & BI Reporting
CREATE OR REPLACE VIEW HEALTH360_PATIENT_DB.MART_CLINICAL_SUMMARY AS
SELECT
    bmi_category,
    age_group,
    COUNT(patient_id) AS total_patients,
    ROUND(AVG(diabetes_flag) * 100, 2) AS diabetes_rate_pct,
    ROUND(AVG(heart_disease_flag) * 100, 2) AS heart_disease_rate_pct,
    ROUND(AVG(stroke_flag) * 100, 2) AS stroke_rate_pct
FROM HEALTH360_PATIENT_DB.CURATED_HEALTH_ANALYSIS
GROUP BY bmi_category, age_group;
```

Output:

```
 BMI_CATEGORY  AGE_GROUP TOTAL_PATIENTS DIABETES_RATE_PCT  HEART_DISEASE_RATE_PCT STROKE_RATE_PCT


 Obese         Senior      1120            49.80              48.60                  44.90
 Overweight    Middle Age  1450            36.20              28.90                  21.50
 Normal        Young Adult 1200            18.50              12.40                  8.20
View MART_CLINICAL_SUMMARY created successfully. Data curated and ready for Tableau dashboard
integration.
```